SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I SARAVANAN.T_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
SARAVANAN.T

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks
1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks
1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Resolution Algorithm – Worked Answers

Five propositional-logic examples with CNF conversion, negated goals, resolution steps, and conclusions.

Method: To prove a goal by resolution, add the negation of the goal to the CNF knowledge base. Resolve complementary literals. Deriving the empty clause (■) means the goal follows from the knowledge base.

Question 1 – Rain and Wet Ground

1. Propositional logic

Let R = “It is raining” and W = “The ground is wet.”

```
Premise 1: R → W
Premise 2: R
Goal: W
```

2. Convert to CNF

R → W is equivalent to ¬R ∨ W. The fact R is already a clause.

```
C1: ¬R ∨ W
C2: R
```

3. Negate the goal

```
C3: ¬W
```

4. Resolution steps

```
Resolve C1 (¬R ∨ W) with C2 (R): C4 = W
Resolve C4 (W) with C3 (¬W): ■
```

Resolution tree: (¬R ∨ W) and R derive W; W and ¬W derive ■.

Conclusion: The empty clause is derived, so the ground is wet.

Question 2 – Student Assignment Submission

1. Propositional logic

Let S = “Rahul submitted the assignment” and M = “Rahul receives internal marks.”

```
Premise 1: S → M
Premise 2: S
Goal: M
```

2. CNF clauses

```
C1: ¬S ∨ M
C2: S
```

3. Negate the goal

```
C3: ¬M
```

4. Resolution steps

```
Resolve C1 (¬S ∨ M) with C2 (S): C4 = M
Resolve C4 (M) with C3 (¬M): ■
```

Conclusion: The empty clause is derived, so Rahul receives internal marks according to the knowledge base.

Question 3 – Library Membership

1. Propositional logic

Let L = “Priya is a library member” and B = “Priya can borrow books.”

```
Premise 1: L → B
Premise 2: L
Goal: B
```

2. CNF clauses

```
C1: ¬L ∨ B
C2: L
```

3. Negate the goal

```
C3: ¬B
```

4. Resolution until contradiction

```
Resolve C1 (¬L ∨ B) with C2 (L): C4 = B
Resolve C4 (B) with C3 (¬B): ■
```

Conclusion: Since ■ is obtained, Priya can borrow books follows from the knowledge base.

Question 4 – Placement Eligibility

1. Propositional logic

Let A = “Arun cleared the aptitude test” and E = “Arun is eligible for placement.”

```
Premise 1: A → E
Premise 2: A
Goal: E
```

2. CNF clauses

```
C1: ¬A ∨ E
C2: A
```

3. Negate the goal

```
C3: ¬E
```

4. Resolution steps and tree

```
C1 (¬A ∨ E) + C2 (A) → C4 (E)
C4 (E) + C3 (¬E) → ■
```

Tree in text form:

```
(¬A ∨ E) A
\ /
E ¬E
\ /
■
```

Conclusion: The empty clause proves that Arun is eligible for placement.

Question 5 – Access Control System

1. Propositional logic

Let P = “The user entered the correct password,” H = “The user is authenticated,” and G = “The user is granted access.”

```
Premise 1: P → H
Premise 2: H → G
Premise 3: P
Goal: G
```

2. Convert implications to CNF

C1: ¬P ∨ H

C2: ¬H ∨ G

C3: P

3. Negate the goal

C4: ¬G

4. Resolution steps

Step 1: Resolve C2 (¬H ∨ G) with C4 (¬G): C5 = ¬H

Step 2: Resolve C1 (¬P ∨ H) with C5 (¬H): C6 = ¬P

Step 3: Resolve C6 (¬P) with C3 (P): ■

5. Resolution tree (text form)

(¬H ∨ G) ¬G

\ /

¬H

(¬P ∨ H) ¬H

\ /

¬P P

\ /

■

Conclusion: The empty clause is derived. Therefore, the user is granted access follows logically from the three premises.

Quick rule to remember: In resolution, choose two clauses containing opposite literals (such as X and ¬X), cancel that pair, and combine any remaining literals. If no literals remain, the result is the empty clause ■.